

Parameters

Server:

- Maximum Budget Q_s
- Period T_s

Task:

- Worst Case Execution Time C_j
- Minimum interarrival Time T_j

Variables

Server:

- Budget c_s
- Server deadline d_s
- Server Utilization/Bandwidth $U_s = Q_s/T_s$

Tasks:

- Hard Real-Time Task τ_j
- Job J_j
- Arrival time r_j
- Task deadline d_j
- f_j = finishing time of job

Rules

- Server is "active" if there exist served job J_i s.t. $r_i \leq t \leq f_i$
Otherwise server is "idle"
- Served jobs are enqueued in a First in First Out queue

Pseudo Code Soft

New Job J_i Arrival:

If [pending jobs]:

enqueue new job

Else if [$U_s < c_s / (d_s - r_i)$]:

$c_s = Q_s$

$d_s = r_i + T_s$

Start job

Budget exhausted:

$c_s = Q_s$

$d_s = d_s + T_s$

Definitions

Server:

- Budget c_s
- Maximum Budget Q_s
- Period T_s
- Server deadline d_{sk}

Tasks:

- Hard Real-Time Task τ_i
- Job J_{ij}
- Arrival time r_{ij}
- Task deadline d_{ij}
- $C_i = \text{WCET}$
- $T_i = \text{minimum interarrival time}$

Pseudo Code Soft

New Job Arrival:

 If [pending jobs]:

 enqueue new job

 Else if [$U_s < c_s / (d_s - r)$]:

$c_s = Q_s$

$d_s = r + T_s$

 Start job

Budget exhausted:

$c_s = Q_s$ (replenish)

$d_s = d_s + T_s$ (reset deadline)

Constant Bandwidth Example

- Should start job be added in the constant bandwidth case?
- Is misaligned a good example?

Constant Bandwidth Example – Part 3

Task 1: ||o^x||x||
Server: 2||2||1||0||

^ = arrival task time delimiter
|| = time delimiter
x = execution
o = idle

Task Parameter Table

Tasks C T
Tau_1 2 3

Server Parameter Table

Q_s 2
T_s 3

Server Variable Table

Budget ?
Server
Deadline ?
Utilization ?

Pseudo Code Soft

New Job J_i Arrival:

 If [pending jobs]:

 enqueue new job

 Else if [$U_s < c_s / (d_s - r_i)$]:

$c_s = Q_s$

$d_s = r + T_s$

 Start job

Budget exhausted:

$c_s = Q_s$

$d_s = d_s + T_s$